

Experiment 1: Case Study

Problem Statement: Choose a real-world case study such as a Movie Ticket Booking System, create user stories, select an appropriate software development process model, identify a suitable programming language, and propose an implementation and evaluation strategy.

Real-World Case Study: Movie Ticket Booking System

1. Functional Requirements

Customer

- Register and login
- Search movies
- View movie details
- Select theatre and show timings
- Choose seats
- Book tickets
- Make online payments
- Track booking status
- Download e-ticket
- Cancel booking
- View booking history
- Submit movie reviews

Admin

- Manage movies
- Manage theatres
- Manage screens
- Update show timings
- View customer details
- Process bookings
- Generate booking reports

2. User Stories

User ID	User Story	Priority
US-01	Register	High
US-02	Login	High
US-03	Search movies	High
US-04	Select seats	High

US-05	Cancel booking	Medium
US-06	Book tickets	High
US-07	Pay online	High
US-08	Download e-ticket	Medium
US-09	Add movies	High
US-10	Manage shows	High
US-11	Reports	Medium
US-12	Manage customers	Medium

3. Selected Software Development Process Model

Agile Scrum

Reasons:

- Requirements change frequently.
- Continuous customer feedback.
- Incremental delivery.
- Faster development cycles.
- Easy bug fixing.
- Suitable for web applications.

Sprint 1 Registration & Login

Sprint 2 Movie Catalog & Search

Sprint 3 Seat Selection

Sprint 4 Booking

Sprint 5 Payment Gateway

Sprint 6 Admin Dashboard

Sprint 7 Testing & Deployment

4. Programming Language Selection

Backend: Java (Spring Boot)

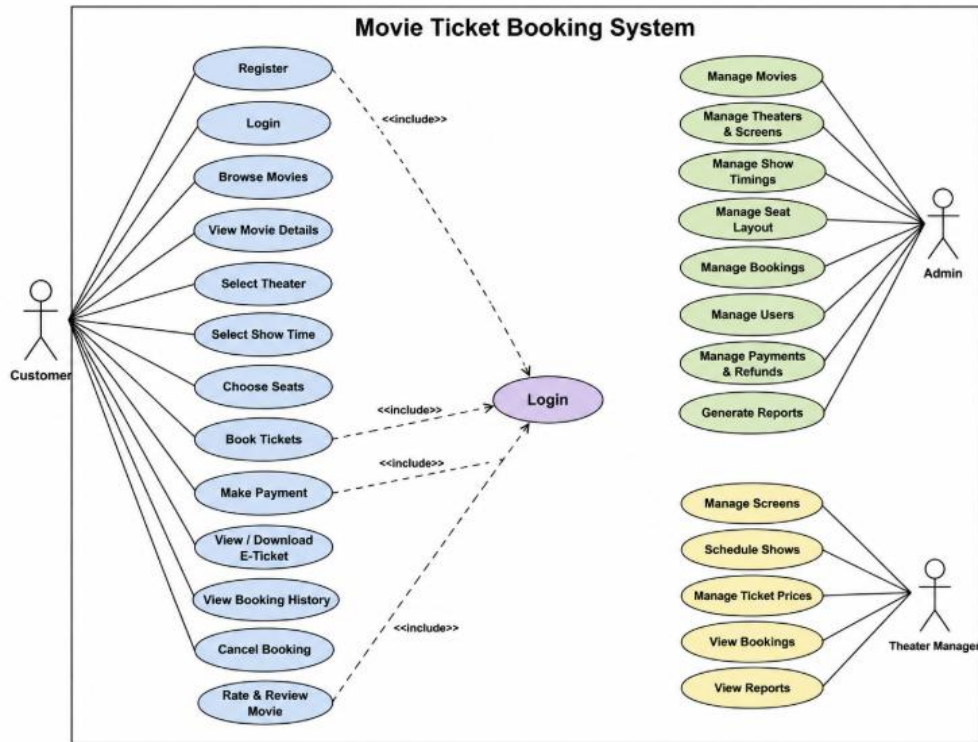
Frontend: HTML5, CSS3, JavaScript, React.js

Database: MySQL

Other Technologies: REST API, JWT Authentication, Razorpay/Stripe, Git, Docker, AWS

5. System Architecture

Customer -> React Frontend -> REST API -> Spring Boot Backend -> Authentication, Booking, Payment, Admin Modules -> MySQL Database



6. Implementation Strategy

Phase 1 Requirement Analysis

Phase 2 System Design

Phase 3 Development

Phase 4 Testing (Unit, Integration, System, UAT)

Phase 5 Deployment using Docker, AWS, MySQL and Nginx.

7. Evaluation Strategy

Functional Accuracy, Performance (JMeter), Security Testing, Reliability, Usability, Scalability, Maintainability.

8. Security Features

JWT Authentication, OAuth2, Multi-Factor Authentication, BCrypt Password Hashing, RBAC, HTTPS/TLS, AES-256 Encryption, SQL Injection Prevention, XSS Protection, CSRF Protection, API Rate Limiting, Secure Cookies, Session Timeout, Seat Locking, QR Ticket Verification, Audit Logs, Intrusion Detection, Encrypted Backups, Disaster Recovery.

9. Expected Outcomes

Secure movie booking platform, fast search, reliable booking, secure payments, instant e-ticket generation, improved customer satisfaction, scalable architecture.